ZOHO

Test Summary

- No. of Sections: 3
- No. of Questions: 171
- Total Duration: 400 min

Section 1 - ZOHO 2

Section Summary

- No. of Questions: 40
- Duration: 200 min

Additional Instructions:

None

Q1. ***Remove palindrome words***

string should contains only the words are not palindrome

Input Format

Input is a string

Output Format

Print the altered string

Sample Input

```
Malayalam is my mother tongue
```

Sample Output

```
is my mother tongue
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q2. **Frequency Sorting**

Given an array of integers arrange them in the descending order of their frequencies

Sample Input

```
11
1 2 3 1 1 1 3 2 4 4 2
```

Sample Output

```
1 1 1 1 2 2 2 3 3 4 4
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q3. 10 + 20 = ________

Q4. ***First maximum and first minimum and so on***

Given an array and arrange it with first maximum and first minimum and second maximum and second minimum and so on without using sorting and second array

Input Format

Input will have size and the values

Output Format

Print the required output

Constraints

1<=size<=1000

Sample Input

```
15
5 15 10 25 55 35 75 45 95 50 70 40 60 90 3
```

Sample Output

```
95 3 90 5 75 10 70 15 60 25 55 35 50 40 45
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q5. **BINARY - DECIMAL**

A positive integer is called Binary-Decimal it contains only 0's and 1's
Sample Input0:
32
Sample Output0:
11 11 10
3
Explanation:
There are many possibilities for representing 32 as a sum of Binary-Decimals
Few possibilities will be
10 + 10 + 1+ 1
Count = 5
11 + 10 + 10 + 1

Count = 4
11+11+10
Count = 3
The Expected output is(11 + 11 + 10) as it has minimum number of Binary-Decimals
(Count – 3)

Sample Input

```
32
```

Sample Output

```
11 11 10
3
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q6. **SNAKE PATTERN**
(DONT use MATRIX)

Input Format

N = 5

Output Format

```
            1 2 3 4 5
         10 9 8 7 6
      11 12 13 14 15
   20 19 18 17 16
21 22 23 24 25
```

Sample Input

```
5
```

Sample Output

```
            1   2   3   4   5
       10   9   8   7   6
   11 12 13 14 15
20 19 18 17 16
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q7. **SPLIT THE STRING ACCORDING TO DICTIONARY**
Given an input string and a dictionary of word,find out if the input string can be segmented into a space separated sequence of dictionary words

Consider the following dictionary
{i , like , ice , cream , icecream }
the input string is ilikeicecream the expected output is
i like icecream

Input Format

N - no of words in dictionary
dictionary of words
input string

Output Format

display the separated input string

Sample Input

```
5
i like ice cream icecream
ilikeicecream
```

Sample Output

```
i like icecream
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q8. **LATIN SQUARE**
write a program to construct a latin square of a given N without using
i) any conditional statements (if - else / ternary operator)
ii) matrix
A latin square is an n X n matrix array filled with n different symbols , each occurring exactly once in each row and exactly once in each column

Sample Input

```
3
```

Sample Output

```
A B C

C A B
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q9. **PATTERN**

Sample Input

```
5
```

Sample Output

```
1
1 1
2 1
1 2 1 1
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q10. **DECOMPRESS THE STRING**

Assume that the given string has enough memory
Don't use any extra space(IN-PLACE)

Sample Input

```
a2b4c6
```

Sample Output

```
aabbbbcccccc
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q11. ***Convert number to words***
range is 0-999

Sample Input

```
234
```

Sample Output

```
two hundred and thirty four
```

Sample Input

```
200
```

Sample Output

```
two hundred only
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q12. ***Spiral pattern***

Sample Input

```
5
```

Sample Output

```
5 5 5 5 5 5 5 5 5
5 4 4 4 4 4 4 4 5
5 4 3 3 3 3 3 4 5
5 4 3 2 2 2 3 4 5
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q13. ***Alternate sort in unsorted array***
(no extra space)

Sample Input

```
9
23 7 8 30 18 12 6 28 16
```

Sample Output

```
23 7 18 12 16 28 8 30 6
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q14. ***Print the string***
note : no extra space
works only on odd length string
if the string is welcome

```
w             e
   e        m
      l   o
        c
      l   o
   e        m
w             e
```

Input Format

Input will be a string

Output Format

Print the string in the given format

Sample Input

```
welcome
```

Sample Output

```
w             e
   e        m
      l   o
        c
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q15. ***Find a Sub string***
Find if a String2 is substring of String1. If it is, return the index of the first occurrence. else return -1.

Sample Input

```
thistest123string
123
```

Sample Output

```
8
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q16. ***Merge two sorted arrays without duplication***
Output is a merged array without duplicates

Input Format

N1 - no of elements in array 1
array elements for array 1
N2 - no of elements in array 2
array elements for array2

Output Format

display the merged array

Sample Input

```
5
1 2 3 6 9
4
2 4 5 10
```

Sample Output

```
1 2 3 4 5 6 9 10
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q17. ***Remove unbalanced parenthesis***
Remove unbalanced parentheses in a given expression.

Sample Input

```
((abc)((de))
```

Sample Output

```
(abc)((de))
```

Sample Input

```
(((ab)
```

Sample Output

```
(ab)
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q18. ***Array with threshold value***
Given an array and a threshold value find the o/p
i/p {5,8,10,13,6,2}
threshold = 3
o/p count = 17
explanation:

Number	parts	counts
5	{3,2}	2
8	{3,3,2}	3
10	{3,3,3,1}	4
13	{3,3,3,3,1}	5
6	{3,3}	2
2	{2}	1

Input Format

N - no of elements in an array
array of elements
threshold value

Output Format

display the count

Sample Input

```
6
5 8 10 13 6 2
3
```

Sample Output

```
17
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q19. ***Triangle Pattern***
Note : Don't use any matrix
N= 7
1 8 14 19 23 26 28
2 9 15 20 24 27
3 10 16 21 25
4 11 17 22
5 12 18
6 13
7

Sample Input

```
5
```

Sample Output

```
1 6 10 13 15
2 7 11 14
3 8 12
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q20. ***Find the different pair***
print the pair which are mismatched in two strings

Input Format

Input will have two strings

Output Format

Print the mismatched pair separated by comma

Sample Input

```
abcdefgh
abdfhjfb
```

Sample Output

```
c,d d,f e,h f,j g,f h,b
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q21. Sliding Window
Given an array of numbers and a window of size k. Print the maximum of numbers inside the window for each step as the window moves from the beginning of the array.

Input Format

Input contains the array size , no of elements and the window size

Output Format

print the maximum of numbers

Constraints

1 <= size <= 1000

Sample Input

```
8
1 3 5 2 1 8 6 9
3
```

Sample Output

```
5 5 5 8 8 9
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q22. In the following line print only the words that are not palindrome
Sample Input0: He did a good deed
Sample Output0: He good
Sample Input1: Malayalam is my mother tongue
Sample Output1: is my mother tongue

Input Format

Input contains the string

Output Format

Print the altered string

Constraints

Should not use extra memory
1 <= length <= 1000

Sample Input

```
He did a good deed
```

Sample Output

```
He good
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q23. ***Sorting Based on no of factors***
To find the factors of the numbers given in an array and to sort the numbers in descending order according to the factors present in it

Input Format

Input contains the array size and the values

Output Format

print the array which is sorted by the factors count

Constraints

1 <= array_size <= 1000

Sample Input

```
5
8 2 3 12 16
```

Sample Output

```
12 16 8 2 3
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q24. ***Odd numbers in a range***
To find the odd numbers in between the range.

Input Format

Input represents two integers start and end range

Output Format

Print the odd numbers separated by space

Constraints

1<= start<=end<=1000000

Sample Input

```
2 15
```

Sample Output

```
3 5 7 9 11 13
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q25. ***Evaluate mathematical expression***
Check whether a given mathematical expression is valid.

Sample Input

```
(a+b)(c+d+e)
```

Sample Output

```
VALID
```

Sample Input

```
(a+b)(c+d)
```

Sample Output

```
VALID
```

Sample Input

```
(a+b))
```

Sample Output

```
INVALID
```

Sample Input

```
(ab+)
```

Sample Output

```
INVALID
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q26. ***Leap Year***
find whether the given year is leap year or not

Input Format

Input will be an integer

Output Format

Print Leap / Non-leap

Sample Input

```
1990
```

Sample Output

```
Non-leap
```

Sample Input

```
2000
```

Sample Output

```
Leap
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q27. ***Number and its occurrence***
Given a array with n elements print the number of occurrences of that number each number in that array. The order of number doesn't matter. You can reorder the elements.

Input Format

N - no of elements
array of elements

Output Format

display number followed by counts

Sample Input

```
10
4 7 18 16 14 16 7 13 10 2
```

Sample Output

```
4-1
7-2
18-1
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q28. ***Reverse a String***
Given a string as input, you have to reverse the string by keeping the punctuation and spaces intact. You have to modify the source string itself without creating another string.

Constraints

1<=string length<=500

Sample Input

```
A man, in the boat says : I see 1-2-3 in the sky
```

Sample Output

```
y kse, ht ni3 21ee sIsy : a sta o-b-e ht nin amA
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q29. ***String pattern***
print the string as the following pattern
(only for odd length string)

Sample Input

```
Hello
```

Sample Output

```
    l
   ll
  llo
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q30. ***WORD REVERSAL USING RECURSIVE***
Using Recursion to reverse the string such as

Input Format

one two three

Output Format

three two one

Constraints

1 <= string length <= 200

Sample Input

```
i love india
```

Sample Output

```
india love i
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q31. ***cyclic number verification***
verify the given number is cyclic or not

Input Format

Num1 num2

Constraints

1<=range<=9999999999

Sample Input

```
12345 45123
```

Sample Output

```
Yes
```

Sample Input

```
12345 54123
```

Sample Output

```
No
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q32. ***Pattern***

Sample Input

Sample Output

4	4444 4334 4334 4444

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q33. ***Reverse and Add until get a palindrome***
Take a number, reverse it and add it to the original number until the obtained number is a palindrome

Constraints

1<=num<=99999999

Sample Input	Sample Output
32	55

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q34. ***Remove Characters***
Given two Strings s1 and s2, remove all the characters from s1 which is present in s2.

Input Format

Input will have two strings

Output Format

Print the string

Constraints

1<= string length <= 200

Sample Input	Sample Output
experience enc	xpri

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q35. **Excel Sheet**
Given a number, convert it into corresponding alphabet.
Input Output
1 A
26 Z
27 AA
676 YZ

Input Format

Input is an integer

Output Format

Print the alphabets

Constraints

`1 <= num <= 4294967295`

Sample Input	Sample Output
26	Z

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q36. **Roman to Decimal**
Given a Roman numeral, find its corresponding decimal value.

Input Format

Input is a string which contains Roman numbers

Output Format

Print the decimal value

Constraints

1<=string_length<100

Sample Input	Sample Output
XLV	45

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q37. ***New Number System***
Form a number system with only 3 and 4. Find the nth number of the number system.
Eg.) The numbers are: 3, 4, 33, 34, 43, 44, 333, 334, 343, 344, 433, 434, 443, 444, 3333, 3334, 3343, 3344, 3433, 3434, 3443, 3444 ....

Input Format

Input will be an integer

Output Format

Print the nth number

Constraints

1<=N<=10000000

Sample Input

```
10
```

Sample Output

```
344
```

Sample Input

```
6743
```

Sample Output

```
434334344333
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q38. ***Add 2 numbers in given Base***
Add two numbers in the given base without converting into base

Input Format

get two numbers and base

Output Format

display the sum

Sample Input

```
1010 11001 2
```

Sample Output

```
100011
```

Sample Input

```
123  13 4
```

Sample Output

```
202
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q39. **Excel Sheet**
Given a number, convert it into corresponding alphabet.
Input Output
1 A
26 Z
27 AA
676 YZ

Input Format

Input is an integer

Output Format

Print the alphabets

Constraints

`1 <= num <= 4294967295`

Sample Input

```
26
```

Sample Output

```
Z
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Q40. **Check if a number is a power of another number**
Given two positive numbers x and y, check if x is a power of y or not.

Examples :

Input: x = 100, y = 5
Output: True

Input: x = 1000, y = 10

Output: True

Input: x = 10001, y = 10
Output: False

Sample Input

```
625 5
```

Sample Output

```
True
```

Sample Input

```
128 5
```

Sample Output

```
False
```

Time Limit: - ms Memory Limit: - kb Code Size: - kb

Section 2 - ZOHO 1

Section Summary

- No. of Questions: 125
- Duration: 100 min

Additional Instructions:
None

Q1.

1. Find the output for the following programs

```
#include<stdio.h>
Void main()
{
int i;
for( i = 1 ; i < 4 ; i++)
{
 switch(i)
 {
   case 1 : printf("%d" , i);break;
   case 2 : printf("%d" , i);break;
   case 3 : printf("%d" , i);break;
 }

}
switch(i)
{
   case 4 : printf("%d" , i);break;
}
} _________
```

Q2.

1. Find the output

```
void main()
{
   int a[10] , i = 0;
   for(i = 0 ; i<10 ; i++)
        a[i] = 9 - i;
   for(i = 0 ; i < 10 ; i++)
        a[i] = a[a[i]];
   for(i = 0 ; i < 10 ; i++)
        printf("%d " , a[i]);
} _________
```

Q3.

1. Find the output

```
void printout(char* pstr)
{
   int iretval = 0;
   if(pstr)
   {
       while(*pstr && *pstr<= '9' && *pstr >= '0')
       {
           iretval = (iretval*10)+(*pstr - '0');
           pstr++;
       }
   }
 printf("%d\n" , iretval);
}
void main()
{
   printout("X32");
   printout("47X74");
} _________
```

Q4.

1. Find the output

```
void main()
{
char *s = "\12345s\n";
```